

WORLD MONITOR

Crisis & Executive Security Platform

Technical Feature Roadmap

100% Free & Open-Source Stack · 4 Phases · 12 Months

Target Users	Base Platform
Crisis Operations Teams Corporate Security Teams Executive Protection Units	TypeScript + Vite + Tauri Vercel Edge + Railway deck.gl + MapLibre

1. Executive Summary

This roadmap defines the technical plan to extend World Monitor from a general-purpose OSINT dashboard into a purpose-built Crisis Team Surveillance & Executive Security platform. All proposed technologies are free and open-source. The roadmap is organized into four sequential phases over 12 months, each delivering independently valuable capabilities that compound on each other.

Design Constraint

Every library, framework, and service added must carry a permissive open-source license (MIT, Apache 2.0, BSD, LGPL, or AGPL). No proprietary SDKs. No paid tiers required for core functionality.

Phase	Theme	Timeline	Key Deliverable
Phase 1	Foundation — Auth, Roles & Alert Dispatch	Months 1–2	Ops-mode access control + outbound alerts
Phase 2	Crisis Command — War Room & Incident Management	Months 3–5	Shared incident state + SITREP generator
Phase 3	Executive Security — Travel & Personal Watch	Months 6–9	Travel briefings + named-entity watchlist
Phase 4	Intelligence Depth — RAG, Voice & Offline	Months 10–12	Voice brief + full offline field mode

2. Full Open-Source Technology Stack

The following technologies extend the existing World Monitor stack. All are free and open-source with no required paid tiers for core functionality.

2.1 Authentication & Access Control

Category	Technology	License	Purpose
Auth	Lucia Auth v3	MIT	Session-based auth framework for TypeScript — works with any database adapter
Auth	Oslo (by Lucia)	MIT	Cryptographic utilities: TOTP, PKCE, JWT, session tokens
2FA	otpauth (npm)	MIT	TOTP/HOTP library for browser + Node — compatible with Authenticator apps
DB (Auth)	Drizzle ORM + libSQL	Apache 2.0	Type-safe ORM; libSQL is open-source fork of SQLite for edge deployments
Session	Upstash Redis (existing)	MIT client	Session storage with automatic TTL expiry — already in stack

2.2 Real-Time Collaboration & War Room

Category	Technology	License	Purpose
WebSocket	Hocuspocus Server	MIT	Y.js WebSocket provider — built for collaborative document/state sync
CRDT	Y.js	MIT	Conflict-free replicated data types — shared incident state across all analysts
Presence	PartyKit (self-hosted)	Apache 2.0	Lightweight presence + room state server; self-hostable on Railway
Pub/Sub	Redis Pub/Sub (Upstash)	MIT client	Event broadcast across edge functions for alert distribution

2.3 Alert Escalation & Outbound Dispatch

Category	Technology	License	Purpose
Email	Nodemailer + SMTP	MIT	Email dispatch via any SMTP server (Postal, Mailhog for dev)
Email Server	Postal (self-hosted)	MIT	Full open-source mail transfer agent — self-hosted on Railway/VPS
SMS	Fonoster	MIT	Open-source Twilio alternative — SMS + voice via SIP/VoIP providers
Webhook	Standard Fetch API	N/A	Outbound Slack/Teams/Discord webhooks — no library needed
PagerDuty alt	Grafana OnCall (self-hosted)	Apache 2.0	Open-source on-call scheduling + escalation policy engine
Queue	BullMQ + Redis	MIT	Job queue for escalation timers and retry logic

Scheduler	node-cron	MIT	Cron-based scheduling for recurring checks and digest emails
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2.4 Document Generation (SITREP / Briefings)

Category	Technology	License	Purpose
Word/DOCX	docx (npm)	MIT	Programmatic Word document generation — already proven in this stack
PDF	Puppeteer (headless Chrome)	Apache 2.0	HTML-to-PDF rendering for country briefs and SITREPs
PDF alt	pdfmake	MIT	Pure-JS PDF generation without headless browser dependency
Templates	Handlebars.js	MIT	Logic-less templating for SITREP HTML shells before PDF conversion
Charts in PDF	Chart.js + canvas	MIT	Server-side chart rendering via node-canvas for embedded PDF charts

2.5 Geofencing & Spatial Triggers

Category	Technology	License	Purpose
Geofencing	@turf/turf (modular)	MIT	Comprehensive geospatial library — point-in-polygon, buffer, intersect
H3 Grid	h3-js	Apache 2.0	Uber's hexagonal indexing — already used for GPS jamming layer
Spatial DB	SQLite + SpatiaLite	LGPL	Lightweight spatial queries for offline geofence evaluation
GeoJSON	geojson-validation	MIT	Validates user-drawn geofence polygons before persistence

2.6 Voice Interface

Category	Technology	License	Purpose
Speech-to-Text	OpenAI Whisper (local)	MIT	State-of-the-art STT — runs fully locally via whisper.cpp or Ollama
Text-to-Speech	Kokoro TTS	Apache 2.0	High-quality neural TTS with 8 voices — runs in browser via ONNX
Browser STT	Web Speech API	W3C Standard	Built-in browser STT — zero dependencies, instant for modern Chrome/Edge
Wake Word	Porcupine (community)	Apache 2.0	Offline wake-word detection — 'Hey Monitor' trigger without server

2.7 Offline & Field Operations

Category	Technology	License	Purpose
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Offline Storage	Dexie.js (IndexedDB)	Apache 2.0	Elegant IndexedDB wrapper — already partially used in the stack
Offline Sync	ElectricSQL	Apache 2.0	Postgres-to-SQLite sync with offline-first CRDT merge on reconnect
Map Tiles	Protomaps (PMTiles)	BSD	Single-file map tile archive — entire world map in one offline file
Compression	fflate	MIT	Fast pure-JS gzip/Brotli for compressing offline data bundles
Service Worker	Workbox (existing)	MIT	Already in stack — extend strategies for intelligence data caching

2.8 Audit, Logging & Observability

Category	Technology	License	Purpose
Audit Log	audit-log-js	MIT	Structured append-only event log with tamper detection
Logging	Pino	MIT	Extremely fast structured JSON logger for Node.js sidecar
Monitoring	OpenTelemetry JS	Apache 2.0	Distributed tracing and metrics — exports to Grafana/Prometheus
Metrics	Prometheus client (prom-client)	Apache 2.0	Metrics collection for alert latency, escalation counts, API health
Dashboards	Grafana (self-hosted)	AGPL-3.0	Operational dashboards for system health — same license as World Monitor

3. Phase 1 — Foundation: Auth, Roles & Alert Dispatch

P1	Authentication, RBAC & Outbound Alerting	Months 1–2 (8 weeks)	START HERE
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Phase 1 converts World Monitor from a single-user monitoring wall into a multi-user operational tool. Without authentication and outbound dispatch, every other feature in this roadmap has limited operational value. These two capabilities are the prerequisite for everything else.

3.1 Role-Based Access Control (RBAC)

Implement a three-tier role model with session-based authentication using Lucia Auth v3 and Drizzle ORM with libSQL (SQLite-compatible, deployable to Railway).

Role	Badge	Permissions
Viewer	Read-only link access	Dashboard read, no alerts, no incident interaction
Analyst	Operations staff	Acknowledge alerts, annotate incidents, trigger deductions, export SITREPs
Duty Manager	Incident commander	Declare/close incidents, escalate alerts manually, manage geofences
Admin	Platform owner	User management, API key config, audit log access, system settings

Implementation Steps

- **Step 1** Install Lucia Auth v3 + Drizzle ORM adapter for libSQL
- **Step 2** Define users table with role enum (viewer/analyst/manager/admin) and team_id for multi-tenant isolation
- **Step 3** Add TOTP 2FA using the oslo library for Analyst and above — QR code provisioning via otpauth
- **Step 4** Gate all existing Vercel edge functions behind a session middleware check using the existing CORS allowlist pattern
- **Step 5** Add invite-link flow: Admin generates time-limited signed token (24h TTL in Redis), recipient sets password on first visit
- **Step 6** Add read-only public share tokens for Viewer role — existing URL state sharing continues to work for anonymous views

3.2 Alert Escalation Chain with Outbound Dispatch

Extend the existing signal aggregator to support user-configured thresholds with a tiered escalation chain. Uses BullMQ for the escalation timer queue and Nodemailer + Fonoster for outbound channels.

Architecture Note

The alert engine runs as a persistent worker on Railway — the same server that hosts the AIS relay. BullMQ jobs are stored in the existing Upstash Redis instance. No new infrastructure required.

Escalation Flow

- L1 (Analyst) — alert fires in UI; 10-minute acknowledgment window via BullMQ delayed job
- L2 (Duty Manager) — if unacknowledged at T+10min, repeat alert + email/Slack notification dispatched
- L3 (Phone/SMS) — if unacknowledged at T+20min, SMS via Fonoster (self-hosted) + optional voice call
- Auto-resolve — if source signal drops below threshold, BullMQ job cancelled and incident auto-closed

User-Configurable Threshold Types

- CII score crosses a user-defined value for a specific country (e.g., Israel > 75)
- Event type occurs within a user-drawn geofence polygon (Phase 1 uses bounding box; full polygon in Phase 2)
- Named keyword detected in RSS/Telegram feeds with user-defined severity floor
- OREF alert fires for any location OR a specific region prefix
- Theater posture elevates to CRITICAL for a selected theater

Outbound Channels — All Open-Source

- Email — Nodemailer via self-hosted Postal SMTP or any third-party SMTP (Gmail, Proton, etc.)
- Slack/Teams/Discord — standard incoming webhook POST, no SDK required
- SMS — Fonoster open-source via any SIP provider (Twilio, Bandwidth, or self-hosted FreeSWITCH)
- Webhook — arbitrary HTTP POST with user-defined URL and custom payload template (Handlebars)

3.3 Phase 1 Feature Summary

Feature	Tech / Library	Effort	Priority
Lucia Auth v3 + RBAC (4 roles)	<i>lucia-auth, drizzle-orm, libSQL</i>	Medium	Critical
TOTP 2FA for Analyst+	<i>oslo, otpauth</i>	Low	High
Multi-user invite system	<i>Redis TTL tokens, Lucia sessions</i>	Low	High
Alert threshold configuration UI	<i>React state + Drizzle persistence</i>	Medium	Critical
BullMQ escalation timer queue	<i>bullmq, ioredis (Upstash)</i>	Medium	Critical
Email dispatch (Nodemailer)	<i>nodemailer, Postal SMTP</i>	Low	Critical
Slack/Teams/Discord webhooks	<i>Fetch API, Handlebars templates</i>	Low	High
SMS via Fonoster (self-hosted)	<i>fonoster/fonoster</i>	High	Medium
Audit log (append-only)	<i>audit-log-js, libSQL</i>	Low	High

4. Phase 2 — Crisis Command: War Room & Incident Management

P2	Collaborative Incident State, SITREP & Geofencing	Months 3–5 (10 weeks)	DEPENDS ON P1
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Phase 2 transforms World Monitor into a true crisis command tool. The key insight is that your existing WebSocket relay infrastructure (Railway) already provides the transport layer — Phase 2 adds Y.js CRDT state on top of it to synchronize incident state across all connected analysts in real time.

4.1 Shared Incident War Room

When a Duty Manager declares an incident, a Y.js document is created on the Hocuspocus server (deployed alongside the Railway relay). All authenticated analysts connecting to that incident share the same pinned map view, active layers, severity level, and annotation layer.

Y.js CRDT Advantage

Y.js uses Conflict-Free Replicated Data Types — two analysts updating the same field simultaneously will always converge to a consistent state without conflicts, with no central locking. This is how Notion, Linear, and Figma handle concurrent editing.

Shared Incident State Fields (Y.js Document Schema)

- incidentId — UUID, declared by Duty Manager, immutable after creation
- status — enum: DECLARED / ACTIVE / MONITORING / CLOSED
- severity — 1–5 integer; changes broadcast to all connected users instantly
- pinnedMapView — { center, zoom, layers[], timeRange } — synced across all war room participants
- annotations — Y.Array of { id, type, coordinates, author, timestamp, text } for map markup
- watchedCountries — Y.Array of country codes elevated to priority monitoring
- timeline — append-only Y.Array of { timestamp, actor, action, detail } for the shift log
- activeAlerts — Y.Map keyed by alertId with acknowledgment status per analyst

Technical Implementation

- Deploy Hocuspocus Server (Y.js WebSocket provider) on Railway — single process alongside existing AIS relay
- Hocuspocus rooms are named incident:{incidentId} — only users with valid session token and matching team_id are admitted
- Existing URL state sharing (?view=, ?zoom=, ?layers=) becomes a read-only projection of the Y.js pinnedMapView document
- Map annotation layer renders Y.Array annotations as a deck.gl ScatterplotLayer + TextLayer on top of existing globe
- Analyst presence indicators (avatar, cursor position on map) via Hocuspocus Awareness protocol — no extra infrastructure

4.2 SITREP Generator

One-click generation of a structured Situation Report that aggregates the current intelligence picture into a formatted Word document or PDF. Uses the existing docx npm library already proven in this stack.

SITREP Document Sections (Auto-Populated)

- Header — incident ID, classification level, timestamp, issuing analyst name, next update due
- Situation Overview — AI-generated narrative (existing World Brief LLM chain) with source citations
- Country Risk Table — CII scores for all watched countries with delta from previous SITREP
- Active Signals — theater posture status, convergence alerts, infrastructure cascade events
- Timeline — pulled from Y.js incident timeline array, formatted chronologically
- Intelligence Gaps — explicit listing of data sources currently offline (existing gap tracker)
- Recommended Actions — analyst free-text field pre-populated with AI deduction output
- Distribution List — pulled from team member list with role indicators

Export Formats

- Word (.docx) — using docx npm library, editable template with classification banner in header
- PDF — Puppeteer headless Chrome render of an HTML Handlebars template for pixel-perfect output
- JSON — structured data export for downstream systems or custom report builders

SITREP Versioning

Each SITREP export is stamped with a version number (SITREP-001, SITREP-002) and stored in libSQL alongside the incident record. Analysts can retrieve any prior version for the post-incident review.

4.3 User-Defined Geofence Triggers

Analysts draw polygons directly on the map to define watch zones. Any event (conflict, protest, military flight, outage) occurring inside the polygon triggers an alert via the Phase 1 escalation chain. Uses `@turf/turf` for polygon math and H3 for efficient spatial indexing.

Geofence Lifecycle

- Draw — MapLibre draw plugin (maplibre-gl-draw, MIT) provides polygon drawing tools on the map
- Name & Configure — analyst names the zone, selects which event types to monitor, sets severity floor, assigns escalation chain
- Persist — geofence GeoJSON stored in libSQL with `team_id` isolation; loaded at session start
- Index — `@turf/bboxPolygon + H3 covers()` used to pre-compute which H3 cells overlap the polygon for O(1) lookup
- Evaluate — on every signal aggregator update cycle, H3 cell of new event checked against pre-computed geofence index
- Alert — if inside: fires BullMQ job → escalation chain (Phase 1)

4.4 Shift Handover & Audit Trail

Structured shift handover log prevents intelligence loss between analyst rotations. The audit trail ensures post-incident accountability.

- Shift start/end timestamps recorded per analyst with overlap window configurable by Duty Manager

- 'Handover Brief' modal prompts outgoing analyst to summarize: what happened, what is developing, what next shift must watch
- Handover notes appended to Y.js incident timeline and separately persisted in libSQL for offline retrieval
- Full audit trail: every alert acknowledgment, SITREP export, geofence change, and user action timestamped and attributed
- Audit log is append-only — individual entries cannot be deleted, only exported

4.5 Phase 2 Feature Summary

Feature	Tech / Library	Effort	Priority
Y.js CRDT shared incident state	<i>yjs, hocuspocus-server</i>	High	Critical
Real-time analyst presence / cursors	<i>Hocuspocus Awareness API</i>	Low	High
Incident declaration + lifecycle	<i>libSQL, BullMQ, Y.js</i>	Medium	Critical
Shared pinned map view sync	<i>Y.js + existing MapLibre state</i>	Medium	Critical
Map annotation layer	<i>maplibre-gl-draw, deck.gl, Y.js</i>	Medium	High
SITREP Word document generator	<i>docx npm, Handlebars, Pino</i>	Medium	Critical
SITREP PDF export	<i>Puppeteer, pdfmake (fallback)</i>	Medium	High
User-drawn geofence polygons	<i>@turf/turf, maplibre-gl-draw</i>	Medium	Critical
H3-indexed geofence evaluation	<i>h3-js (existing)</i>	Low	High
Shift handover log	<i>Y.js timeline, libSQL persistence</i>	Low	High
Append-only audit trail	<i>audit-log-js, libSQL</i>	Low	High
Incident timeline D3 chart	<i>D3.js (existing)</i>	Low	Medium

5. Phase 3 — Executive Security: Travel Briefs & Personal Watch

P3	Travel Briefings, Named Watchlist & Digital Footprint	Months 6–9 (14 weeks)	DEPENDS ON P1+P2
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Phase 3 adds the executive-facing features that transform World Monitor into a corporate security platform. These features consume the same underlying intelligence data but present it through a lens of specific people, specific itineraries, and specific organizational assets rather than global awareness.

5.1 Executive Travel Risk Briefing Generator

Security teams input an executive's itinerary and receive a tailored travel risk brief for each destination. Every data source already exists in the platform — this is a UX packaging and aggregation problem, not a data acquisition problem.

Briefing Input Fields

- Executive profile (name, role — no biometric data stored, only for document labeling)
- Itinerary: origin city, list of { destination, arrival date, departure date, accommodation area (optional) }
- Output format: DOCX, PDF, or in-app briefing card

Per-Destination Intelligence Sections

- Country Risk — current CII score + component breakdown + 30-day trend, government travel advisory levels from all 4 sources (US, UK, AU, NZ)
- Active Incidents — protests, conflicts, and significant events within 150km of destination city center in the travel window
- Infrastructure Risk — airport operational status (NOTAM + AviationStack), undersea cable health for communications, internet outage history
- Ground Transport — CII scores and event density along major routes from airport to city center using corridor analysis
- Medical & Consular — nearest hospitals (OpenStreetMap Overpass API — free) + embassy emergency contacts (static dataset, offline-capable)
- AI Risk Narrative — existing country brief AI chain generates a 3-paragraph executive summary calibrated to travel context
- Emergency Contacts — local emergency number, embassy 24h line, corporate security ops center number (user-configured)

Technical Architecture

- New proto service: executive/v1 with GetTravelBrief RPC — takes ExecItinerary proto message, returns TravelBriefReport
- Briefing generation runs as a BullMQ job (async, since PDF render can take 5–10s) with polling endpoint for status
- DOCX template uses docx npm with classification banner, corporate logo placeholder, and executive name in header
- Corridor analysis uses @turf/lineString + @turf/buffer (50km) + spatial intersection with protest/conflict event dataset
- OpenStreetMap Overpass API (free, no key required) queried for hospitals within 20km of accommodation coordinates

5.2 Named Person & Asset Watchlist

Security teams register executive names, corporate assets, and sensitive locations. The watchlist continuously scans RSS feeds, Telegram OSINT, and trending keywords for mentions, firing alerts via the Phase 1 escalation chain.

Privacy Architecture

Watchlist entries are stored encrypted (AES-256-GCM via the WebCrypto API) in libSQL with the encryption key held in the OS keychain on desktop, or a Duty Manager password-derived key on web. No plaintext names are ever stored in Redis or logs.

Watchlist Entry Types

- Person — full name + aliases (e.g., 'John Smith', 'J. Smith', 'CEO of Acme Corp')
- Organization — company name + known aliases + ticker symbols
- Location — named address or coordinates with configurable radius (replaces generic geofence for specific assets)
- Domain / IP — for monitoring corporate infrastructure mentions in threat intel feeds

Scanning Pipeline

- RSS scan — after each feed parse, tokenized titles checked against watchlist using existing entity index word-boundary regex pattern
- Telegram scan — 27 OSINT channels already ingested; watchlist terms added to the topic classifier at query time
- Trending keyword scan — watchlist terms added as monitored keywords with a lower spike threshold (1x vs 3x for general terms)
- Threat intel scan — Feodo Tracker, URLhaus, OTX feeds scanned for watchlisted domains and IPs
- Dark web news (via RSS) — optional: add Recorded Future community feeds or IntelligenceX RSS for advanced users

Alert Output

- Each match surfaces in a dedicated Watchlist panel with the matched term highlighted, source, timestamp, and snippet
- Severity scored by source tier (Tier 1 wire service mention = higher severity than Tier 4 blog)
- Pattern: if 3+ sources mention the same watchlist entry within 2 hours, escalate to Duty Manager automatically

5.3 Safe Corridor & Route Risk Overlay

Given an origin and destination within a city or between cities, overlay the route on the map with risk indicators derived from existing event data. Designed for ground transport planning in high-risk environments.

- Route input: free-text origin + destination geocoded via Nominatim (OpenStreetMap, free) or coordinate paste
- Route geometry: OSRM (Open Source Routing Machine — free, self-hostable on Railway) computes the road network path
- Risk overlay: @turf/buffer creates a 10km corridor around the route; existing protest, conflict, and military events within the corridor are surfaced

- CII contribution: corridor passes through multiple countries — each segment's risk is weighted by that country's CII
- Output: route rendered on map as a colored LineLayer (green/amber/red by segment risk), with event markers along the corridor
- Alternative routes: OSRM can return 3 alternatives — security team selects lowest-risk path

5.4 RAG-Powered Security Brief Search

Extend the existing Headline Memory (RAG) system to support security-specific semantic search queries across the full indexed headline archive. 'What threats to pharmaceutical executives have been reported in Southeast Asia this month?' becomes an answerable query against the local IndexedDB vector store.

- New search interface in the Cmd+K palette: prefix query with @ to invoke the RAG search mode
- Queries embedded via existing all-MiniLM-L6-v2 ONNX model — no new model required
- Results ranked by cosine similarity with time-decay weighting (recent headlines scored higher)
- Results passed as context to the existing AI Deduction panel for synthesized response — builds on existing pipeline
- Search history persisted in IndexedDB (encrypted if watchlist encryption is active)

5.5 Phase 3 Feature Summary

Feature	Tech / Library	Effort	Priority
Travel risk briefing generator	OSRM, @turf/turf, docx, Puppeteer	High	Critical
Per-destination CII + advisory pull	Existing CII + advisory stack	Low	Critical
Ground corridor risk analysis	@turf/buffer, @turf/lineString	Medium	High
OpenStreetMap hospital/embassy lookup	Overpass API (free, no key)	Low	High
Named entity watchlist (encrypted)	WebCrypto AES-256, libSQL, Lucia	Medium	Critical
Watchlist RSS + Telegram scan	Extend existing entity index	Medium	Critical
Watchlist alert escalation	BullMQ (Phase 1 chain)	Low	Critical
OSRM route risk overlay	OSRM self-hosted, @turf/buffer	High	High
RAG security query interface	Extend existing Headline Memory	Low	High
Executive profile management UI	React, Lucia auth, libSQL	Medium	High
Briefing version history	libSQL, append-only pattern	Low	Medium
Offline embassy contact directory	Static JSON, service worker cache	Low	High

6. Phase 4 — Intelligence Depth: Voice, RAG & Full Offline

P4	Voice Interface, Advanced RAG & Air-Gapped Field Mode	Months 10–12 (10 weeks)	DEPENDS ON P1–P3
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Phase 4 completes the platform by adding the capabilities that differentiate it from any commercial product: a fully local voice interface, advanced retrieval-augmented generation, and an air-gapped field mode that functions with zero network connectivity.

6.1 Voice-to-Brief Interface

Analysts speak queries to the dashboard and receive synthesized audio responses. Runs entirely locally on desktop using Whisper (STT) and Kokoro TTS — no audio leaves the device.

Architecture

- Wake word detection — Porcupine (Apache 2.0) runs in a background thread listening for 'Hey Monitor'; on desktop Tauri spawns it as a sidecar process
- Speech-to-Text — on wake, browser captures audio via MediaRecorder API; sent to local whisper.cpp HTTP server (spawned by Tauri sidecar) for transcription
- Intent routing — transcribed text parsed for intent: 'brief [country]' → country brief, 'deduction [query]' → AI deduction, 'status' → risk score
- Response generation — existing LLM chain (Ollama → Groq → OpenRouter fallback) generates response text
- Text-to-Speech — Kokoro TTS ONNX model runs in a dedicated Web Worker; audio streamed back as Web Audio API output
- Fallback — browser Web Speech API used as lightweight fallback when whisper.cpp sidecar is unavailable

Supported Voice Commands

- 'Brief [country name]' — reads the AI-generated country brief narrative aloud
- 'Status' — reads the current Strategic Risk Score and top-3 CII countries
- 'Alert [analyst/team]' — voice-triggers the escalation chain for current highest-priority signal
- 'Deduction: [free-text question]' — submits to AI Deduction panel and reads response
- 'Watchlist [name] add/remove' — modifies the watchlist without keyboard interaction

6.2 Advanced RAG with Cross-Source Retrieval

Extend Headline Memory beyond RSS headlines to index all intelligence sources — Telegram OSINT messages, SITREP documents, country briefs, and analyst annotations — into a unified semantic search layer.

Expanded Index Sources

- Telegram OSINT messages — 27 channels × 200-item buffer = up to 5,400 additional vectors; same all-MiniLM model
- SITREP documents — on export, each SITREP is chunked (512 tokens, 64-token overlap) and embedded — enables 'find all SITREPs mentioning X'
- Analyst annotations — war room map annotations indexed for semantic search across past incidents

- Country briefs — AI-generated narrative sections indexed for cross-country pattern detection

Vector Store Scaling

- Current cap is 5,000 vectors (Headline Memory); expand to 50,000 with tiered LRU eviction by source priority
- Approximate nearest-neighbor search via HNSW algorithm (hnswlib-node, Apache 2.0) for sub-50ms search at 50K vectors
- Hybrid retrieval: dense vector search (semantic) + sparse keyword BM25 (wink-bm25-text-search, MIT) combined with RRF fusion

6.3 Air-Gapped Field Mode

A pre-configured offline bundle that field security officers can deploy on a laptop with zero network connectivity. Covers the core intelligence surfaces needed for on-the-ground operations.

Target Use Case

An executive protection officer traveling to a high-risk location. Before departure, they sync the offline bundle. On the ground with no reliable internet, they have full map browsing, recent intelligence, country briefs, travel advisories, emergency contacts, and local RAG search — all offline.

Offline Bundle Contents

- PMTiles map archive — offline vector tile map of selected region(s); bundled at departure for the travel destination
- Intelligence snapshot — last 48 hours of RSS, Telegram, and signal aggregator state serialized to IndexedDB + compressed JSON
- Country briefs — pre-rendered HTML + JSON for all countries in travel region, with AI narratives generated before departure
- Travel advisories — full advisory text from all 4 governments for selected countries, cached with 24h refresh
- Emergency contacts — embassy numbers, hospitals, insurer hotlines — static JSON, always available
- Watchlist local scan — offline watchlist scanning runs against the bundled RSS snapshot without network
- RAG vector index — the indexed vector store exported from desktop to field laptop via encrypted USB transfer

Sync Protocol (Pre-Departure)

- 'Sync for travel' action in desktop app: prompts for destination countries and date range
- Background job fetches and pre-renders all required data, reporting progress (1/7 sources complete, etc.)
- Bundle written to encrypted archive (fflate compression + WebCrypto AES-256) for secure transfer
- On field device: bundle imported, decrypted, and loaded into IndexedDB — no server connection ever required
- On return: delta sync uploads analyst annotations and shift notes to the main platform for incident record

6.4 Push Notifications (PWA + Desktop)

System-level push notifications for critical alerts, delivering escalation notifications even when the dashboard is not the focused browser tab or window.

- Web Push — standard W3C Push API via service worker; self-hosted push server using web-push npm library (MIT)
- Desktop — Tauri native notification API (no external service); OS-level notification with action buttons (Acknowledge / Escalate)
- No third-party push service required — VAPID keys self-generated; push subscriptions stored in libSQL
- Notification payload includes: alert type, country/region, severity, and deep link back to the triggering signal in the dashboard

6.5 Phase 4 Feature Summary

Feature	Tech / Library	Effort	Priority
Whisper STT local sidecar	<i>whisper.cpp HTTP server, Tauri sidecar</i>	High	High
Kokoro TTS in Web Worker	<i>kokoro-js ONNX, Web Audio API</i>	Medium	High
Porcupine wake word detection	<i>porcupine-web (Apache 2.0)</i>	Medium	Medium
Voice command intent router	<i>Regex + existing LLM chain</i>	Low	High
Expanded RAG index (50K vectors)	<i>hnswlib-node, extended Dexie schema</i>	Medium	High
BM25 hybrid retrieval fusion	<i>wink-bm25-text-search, RRF algorithm</i>	Medium	High
SITREP + annotation indexing	<i>Extend headline memory ingestion</i>	Low	Medium
PMTiles offline map bundle	<i>protomaps/PMTiles, MapLibre offline</i>	High	Critical
Pre-departure intelligence sync	<i>fflate, WebCrypto, BullMQ export job</i>	High	Critical
Offline watchlist scan	<i>Extend local watchlist engine</i>	Medium	High
Self-hosted Web Push server	<i>web-push (VAPID), service worker</i>	Medium	High
Tauri native notifications	<i>Tauri notification plugin (existing)</i>	Low	High

7. Implementation Timeline

Feature Area	M1-2	M3	M4-5	M6-7	M8-9	M10-11	M12
Auth + RBAC							
Alert Escalation							
War Room (Y.js)							
SITREP Generator							
Geofence Triggers							
Travel Briefings							
Watchlist (Person)							
Route Risk Overlay							
Voice Interface							
Advanced RAG							
Field Offline Mode							
Push Notifications							

Dependency Note

Phases must be completed in order. Phase 2 requires the authentication system from Phase 1. Phase 3 requires the alert escalation chain from Phase 1. Phase 4 extends capabilities built in all prior phases. Each phase is independently testable and deployable.

8. Deployment Architecture

The extended platform runs across the same three deployment targets as the existing World Monitor — no new infrastructure categories are introduced. New services are co-located with existing ones to minimize operational surface.

8.1 Updated Deployment Map

Platform	Existing Services	New in Roadmap
Vercel Edge	60+ edge functions, SPA, CDN, Redis cache	Auth middleware, RBAC session check, Travel briefing API, Watchlist scan API, Web Push server
Railway	AIS relay, Telegram MTPROTO, OREF proxy, Polymarket proxy	Hocuspocus war room server, BullMQ alert worker, Postal SMTP relay, OSRM routing engine, Whisper.cpp HTTP server
Tauri Desktop	Local sidecar, OS keychain, 60+ local API handlers	Porcupine wake word sidecar, Kokoro TTS Web Worker, PMTiles offline map, Encrypted intelligence bundle
libSQL (Railway)	N/A (Redis only previously)	Users, sessions, teams, incidents, geofences, watchlists, SITREPs, audit log

8.2 Data Storage Split

The new platform uses a deliberate split between Redis (ephemeral, operational data) and libSQL (persistent, relational data). No single point of failure for either intelligence data or organizational data.

- Redis (Upstash, existing) — signal cache, AI result cache, session tokens, BullMQ job store, in-flight request dedup
- libSQL (new, Railway) — users, teams, roles, incidents, geofences, watchlists, SITREPs, audit log, alert rules, shift logs
- IndexedDB (client, existing+extended) — vector store (RAG), offline intelligence bundle, historical playback snapshots
- OS Keychain (Tauri, existing+extended) — API keys, watchlist encryption key, offline bundle decryption key

9. Security Additions

The following security measures are added to the existing World Monitor security model specifically for the crisis and executive security use cases.

Layer	Mechanism	Phase
Multi-user session auth	Lucia Auth v3 + libSQL — server-side sessions, no JWT stored in localStorage	Phase 1
TOTP 2FA	oslo + otpauth — TOTP enforced for Analyst role and above	Phase 1

Watchlist encryption	WebCrypto AES-256-GCM — names encrypted at rest, key in OS keychain	Phase 3
Offline bundle encryption	WebCrypto AES-256 + deflate — bundle encrypted before USB/transfer	Phase 4
Hocuspocus auth	Session token verified before room admission — team_id isolation	Phase 2
Audit log integrity	HMAC signature on each audit entry — tampering detectable	Phase 2
OSRM SSRF protection	Self-hosted OSRM on Railway — internal network call, no external routing API	Phase 3
STT audio privacy	Whisper.cpp runs locally — no audio leaves the device	Phase 4

10. Open-Source License Compliance

All proposed technologies are compatible with the existing AGPL-3.0 license of World Monitor. Permissive licenses (MIT, Apache 2.0, BSD) impose no additional obligations. LGPL libraries are used as external dependencies and do not require source disclosure. The AGPL obligations (source disclosure for network-accessible deployments) already apply to the existing codebase.

Library	License	Compatibility Note
Lucia Auth v3	MIT	No restrictions — full commercial and AGPL use permitted
Y.js + Hocuspocus	MIT	No restrictions — widely used in production collaborative tools
BulMQ	MIT	No restrictions
@turf/turf	MIT	No restrictions — modular import keeps bundle lean
Drizzle ORM	Apache 2.0	Compatible with AGPL — no copyleft obligations as a dependency
OSRM	BSD-2-Clause	Compatible — self-hosted, no API fee, no license fee
Grafana OnCall	Apache 2.0	Compatible with AGPL — self-hosted only
Postal SMTP	MIT	No restrictions — self-hosted mail server
Kokoro TTS	Apache 2.0	Compatible — model weights are also Apache 2.0
OpenAI Whisper (whisper.cpp)	MIT	MIT — the C++ inference engine; original model weights are also MIT
Spatialite	LGPL	Used as external linked library — no source disclosure required
PMTiles / Protomaps	BSD	No restrictions
hnsplib-node	Apache 2.0	Compatible with AGPL

Appendix — Quick-Start Commands by Phase

Phase 1 — New Dependencies

```
npm install lucia @lucia-auth/adapters-drizzle drizzle-orm @libsql/client oslo otpauth bullmq  
nodemailer handlebars audit-log-js
```

Phase 2 — New Dependencies

```
npm install yjs @hocuspocus/server @hocuspocus/provider @turf/turf maplibre-gl-draw puppeteer  
pdfmake pino
```

Phase 3 — New Dependencies

```
npm install @turf/buffer @turf/line-string geojson-validation wink-bm25-text-search
```

Phase 4 — New Dependencies

```
npm install hnsplib-node @picovoice/porcupine-web kokoro-js web-push fflate protomaps-leaflet
```

Railway — New Services to Deploy

```
# Alongside existing ais-relay.cjs node services/hocuspocus-server.cjs # War room sync node  
services/bullmq-worker.cjs # Alert escalation node services/postal-smtp.cjs #  
Email dispatch docker run -p 5000:5000 osrm/osrm-backend # Routing (or Railway Docker)
```